

Modul 09 Basis Data
PRAKTIKUM BASIS DATA
Scripting Procedure And Function



DISUSUN OLEH:
Davis Arvaputra Dwiansyah (103122400034)

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
DIREKTORAT KAMPUS PURWOKERTO
UNIVERSITAS TELKOM
2026

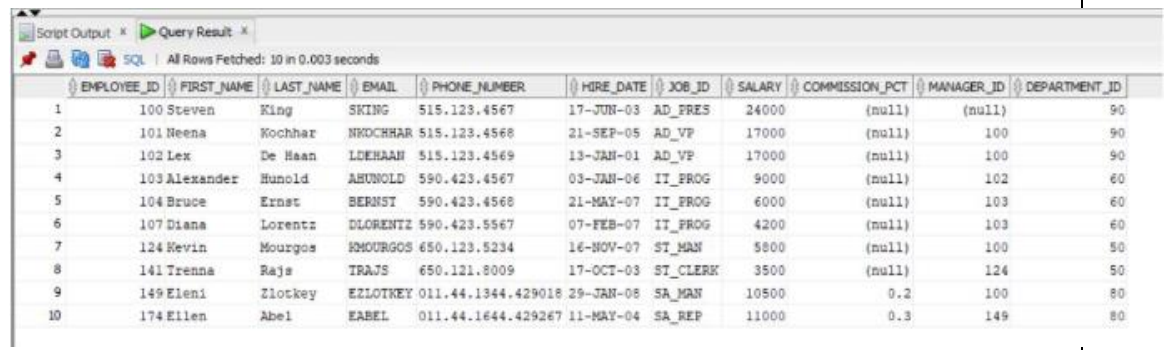
Jurnal

1,2. create table employees as select * from hremployees. select * from employees

Jawaban:

create table employees as select * from hremployees Perintah ini digunakan untuk membuat tabel baru bernama employees dengan cara menyalin seluruh struktur dan data dari tabel hremployees. dan juga select * from employees ini digunakan untuk menampilkan seluruh data yang ada pada tabel employees.

Query & Input



The screenshot shows a SQL query result window with a table containing 10 rows of employee data. The columns are: EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, SALARY, COMMISSION_PCT, MANAGER_ID, and DEPARTMENT_ID. The data is as follows:

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
5	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
6	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
7	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
8	Trenna	Rajs	TTRAJS	650.121.8009	17-OCT-03	ST_CLERK	3500	(null)	124	50
9	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
10	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149	80

3.

```
1 CREATE OR REPLACE PROCEDURE increase_salary (  
2 department_id_in IN employees.department_id%TYPE,  
3 increase_pct_in IN NUMBER)  
4 IS  
5 BEGIN  
6 FOR employee_rec  
7 IN (SELECT employee_id  
8 FROM employees  
9 WHERE department_id = increase_salary.department_id_in)  
10 LOOP  
11 UPDATE employees emp  
12 SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in  
13 WHERE emp.employee_id = employee_rec.employee_id;  
14 DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);  
15 END LOOP;  
16 END increase_salary;  
17
```

BEGIN increase_salary (90, .50); ROLLBACK; END; / select * from employees where department_id = 90

Analisis :

Procedure increase_salary dibuat untuk menaikkan gaji karyawan berdasarkan:

- department_id
- persentase kenaikan gaji

Procedure ini menggunakan perulangan (FOR LOOP) untuk:

- Mengambil setiap employee_id pada department tertentu
- Mengupdate gaji masing-masing karyawan

BEGIN

increase_salary (90, .50);

ROLLBACK;

END;

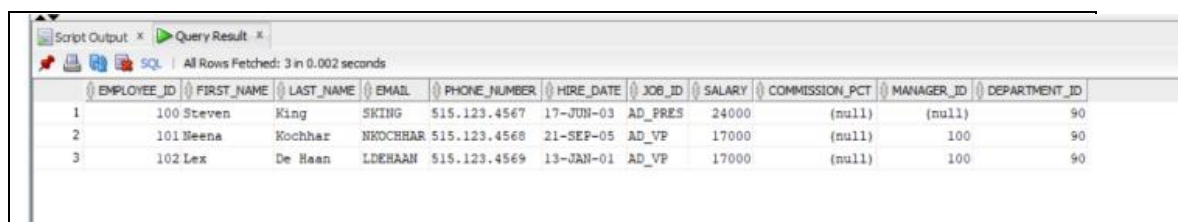
/

- Procedure dijalankan untuk menaikkan gaji karyawan di department 90 sebesar 50% , Namun setelah itu dilakukan ROLLBACK

Query

```
CREATE OR REPLACE PROCEDURE increase_salary (  
    department_id_in    IN employees.department_id%TYPE,  
    increase_pct_in     IN NUMBER  
)  
IS  
BEGIN  
    FOR employee_rec IN (  
        SELECT employee_id  
        FROM employees  
        WHERE department_id = increase_salary.department_id_in  
    )  
    LOOP  
        UPDATE employees emp  
        SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pc  
        WHERE emp.employee_id = employee_rec.employee_id;  
  
        DBMS_OUTPUT.PUT_LINE('Updated ' || SQL%ROWCOUNT);  
    END LOOP;  
END increase_salary;  
/  
  
select * from employees;
```

Output



Script Output: x Query Result: x

SQL All Rows Fetched: 3 in 0.002 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90

3. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.
Jawaban :

Analisis :

Jika perintah ROLLBACK dihapus, maka perubahan yang dilakukan oleh procedure tidak dibatalkan

Query

```
BEGIN
    increase_salary(90, 0.50);
END;
/
SELECT * FROM employees WHERE department_id = 90;
```

Output

Script Output x Query Result x

All Rows Fetched: 3 in 0.003 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPART
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	36000	(null)	(null)	
2	101	Neena	Kochhar	NKOCHHR	515.123.4568	21-SEP-05	AD_VP	25500	(null)	100	
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	25500	(null)	100	